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PERFORMANCE SPECIFICATION

TANK AND PUMP UNIT, FUEL DISPENSING, FOR TRUCK MOUNTING, AND SINGLE TANK, FUEL DISPENSING, 525 GALLON, TRAILER MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Abstract. This specification covers a bottom loading, fuel dispensing, double tank and pump unit for truck mounting, and a single tank for trailer mounting.

1.2 Classification.

1.2.1 Type. The units shall be of the following types as specified (see 6.2):

Type I	Tank and pump unit for truck mounting, fuel dispensing.
Type II	Single tank for trailer mounting, 525 gallons, fuel dispensing.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments, recommendations, additions, deletions, and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-D/210, Warren MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-52030 – Nozzles, Fuel and Oil Servicing, Non-automatic Shutoff and Nozzles, Fuel Servicing, Automatic Shutoff

A-A-59326 – Coupling Halves, Quick-Disconnect Cam-Locking Type

DEPARTMENT OF DEFENSE

MIL-PRF-370 – Hose and Hose Assemblies, Nonmetallic: Elastomeric, Liquid Fuel

MIL-PRF-52109 – Pumping Assemblies, Portable, Bulk Transfer, Fuel and Water Pumping Service

MIL-PRF-52666 – Filter-Separators, Liquid fuel

MIL-PRF-62122 – Cable Assembly, Inter-Vehicle Power: Plug, Receptacle, and Adapter

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-209 – Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment

MIL-STD-462 – Measurement of Electromagnetic Interference Characteristics, Test Method Standard For

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Regulations. The following documents form a part of this document to the extent specified herein.

CODE OF FEDERAL REGULATIONS

49 CFR 178.346 – Specification DOT 406; cargo tank motor vehicle

(Application for copies should be addressed to the Information Center, Office of Hazardous Materials Standards, Research and Special Programs Administration, U.S. Department of Transportation, 400 Seventh Street, SW, Washington, D.C. 20590-0001.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG-1 – Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 N. 17th Street, Suite 1847, Rosslyn, VA 22209.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 – National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, One Batterymarch Park, Quincy, MA 02269-9101.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The type I tank and pump unit shall consist of two tanks and a pump unit, be truck mountable, and dispense liquid petroleum fuels from the two tanks. The pump unit shall consist of a 50 GPM pump assembly, filter/separator, dispensing assembly, and manifold. The type II single tank and manifold shall be trailer mountable and dispense liquid petroleum fuels. The tank(s) shall be internally baffled, low profile, 525-gallon capacity with bottom loading.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Physical characteristics.

3.3.1 Type I. The type I unit shall consist of two tanks, a pumping unit, and manifold, to be mounted on the flatbed of a 5 ton, 6 x 6, cargo truck (see 3.6.1). The tanks shall have a control lever to manually open and close the valves (see 3.3.4) from the ground, at the operator's position. The hoses shall be elastomeric, non-collapsible, reinforced, 2.00-inch (ID) hose conforming to MIL-PRF-370. The hoses shall connect from the bottom-loading valve on the tanks to the pump unit’s manifold connections.

3.3.2 Type II. The type II unit shall consist of one tank and manifold to be trailer mounted. The tank shall have a control lever to manually open and close the valve (see 3.3.4) from the ground, at the operator's position. The hoses shall be elastomeric, non-collapsible, reinforced, 2.00-inch (ID) hose, conforming to MIL-PRF-370. The hose shall connect from the bottom-loading valve on the tank to the manifold connection.

3.3.3 Tank. The tank shall be an internally baffled, low profile, 525-gallon capacity with bottom loading. Components shall include a bottom-loading valve, an access port, an access/egress hole, a fill tube, pilot valve(s), and level sensor(s). The maximum weight of one (1) tank, empty, shall be 500 pounds. The tank shall conform to the Code of Federal Regulations (CFR) Title 49 178.346, except where noted.

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3.3.3.1 Dimensions. The assembled tank shall not be larger than 72.00-inches x 56.00-inches x 44.00-inches (high).

3.3.3.2 Capacity. A device, such as a calibrated dipstick, shall be provided for estimating the volume, in gallons and liters, of the fluid stored in the tank. Provisions shall be made for securing the device on the unit when not in use.

3.3.4 Bottom loading valve assembly. The tank shall have a bottom loading, mechanically operated internal shutoff valve to function as a fuel inlet when bottom-loading and a fuel outlet when offloading. The valve shall automatically open when loading fuel pressure greater than 5.0 psi is applied. A separate manually operated system for controlling the bottom-loading valve shall be installed. The loading fuel pressure range for the bottom-loading valve shall be 5.0 psi to 150.0 psi. The valve shall be fitted with a 90° elbow and a 2-inch, male, quick-disconnect coupling conforming to A-A-59326.

3.3.5 Fuel sensor cutoff. The type I tank and pump unit shall have a level sensor and primary pilot valve. When specified (see 6.2), a secondary pilot valve and a second level sensor shall be provided.

3.3.5.1 Level sensor. The sensor shall operate within the tank for sensing of the fuel level. The sensor shall detect the pressure difference between submerged and emerged conditions. The pressure change shall activate a remote, pilot operated valve, causing flow to stop. The sensor shall have no moving parts. The sensor shall have a “fail safe” system that will signal the main shutoff valve to close when the nozzle flow is interrupted. Operating pressure for the sensor shall be 0 to 150 psi.

3.3.5.2 Primary pilot valve. A primary pilot valve shall be provided for attachment to a remote level sensor. When the rising fuel level reaches the sensor, the pilot valve shall actuate and close the main valve shutting off the fuel flow.

3.3.5.3 Secondary pilot valve. Provisions shall be made for the addition of a secondary pilot valve, for attachment to a second level sensor. When implemented, the flow shall be reduced when the rising fuel level reaches the primary sensor. When the rising fuel level reaches the second level sensor the secondary pilot valve shall actuate, closing the main valve for a positive shutoff of fuel flow.

3.3.6 Tank access.

3.3.6.1 Port. The tank shall have an access port that can be removed to gauge the fuel tank and to pump fuel from the fuel tank. The diameter of the port shall not be less than 2.25-inches.

3.3.6.2 Access hole. The tank shall have an access hole on the top of the tank conforming to the Code of Federal Regulations (CFR) Title 49 178.346, except the access hole shall be an ellipse not less than 11.50-inches x 16.50-inches.

3.3.6.2.1 Access hole cover. The tank shall be equipped with an access hole cover that is provided with a fill hole, cover and tube. The entire access hole cover shall be removable for access to the interior of the tank. Securing of the access hole cover shall be accomplished manually by reaching through the fill tube.

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3.3.6.2.2 Fill hole, cover, and tube. The access hole cover shall be fitted with a minimum 9.50-inch inside diameter fill hole. The fill hole shall have a cover that shall open to provide access to a fill tube that minimizes static charge buildup during top loading. The fill tube shall allow the fuel tank to be filled through the top and shall extend to within 8.00-inches from the bottom of the tank. The fill cover shall be fitted with a fuse plug and vent, and shall have provisions for the user to add an anti-theft padlock.

3.3.6.2.3 Fuse plug and vent. The fill plug shall have a melting point of 170 °F to 190 °F. The fuse plug shall have a discharge rate of 51,500 ft^3/hour at 5.0 psig. The vent shall be capable of opening under a pressure of 0.3 to 5.0 psig and a vacuum pressure of 2.3 to 11.5 oz/in^2. The capacity of the vent shall meet the requirements of Department of Transportation (DOT) 406 and, at a minimum, be rated at 2,880 ft^3/hour at 1.0 psig, and 27,500 ft^3/hour at 5.0 psig.

3.3.7 Manifolds. The manifold shall connect the tank(s), pump assembly, and fuel transfer operations by use of quick-disconnect couplings and hose and hose assemblies (MIL-PRF-370). The type I unit's manifold shall be integrated into the pumping unit. The type II manifold shall have provisions for mounting onto the trailer decking. The manifolds shall have ports as described in Table I.

TABLE I. Manifold ports.

Port connection	Number of ports		Description
	Type I	Type II	
D-1 single point refueling (SPR)	1	NA	2.50-inch, pressurized fuel transfer, single point refueling, adapter, receptacle, and cap, for use with a D-1, SPR, nozzle.
Hose refueling	1	1	2.50-inch, male, quick-disconnect coupling, with cap.
Pump assembly	1	NA	2.00-inch, male, quick-disconnect coupling.
Bottom loading valve	2	1	2.00-inch, female, quick-disconnect couplings.

3.3.8 Pump unit. Type I shall have a pump unit not greater than 72.00-inches x 27.00-inches x 44.00-inches (high). The pump unit shall consist of a pumping assembly, filter/separator, manifold, and dispensing equipment. The pump unit shall be skid mounted and fitted for handling by forklift. The maximum weight of the pump unit shall be 1000 pounds. Pockets shall be adequate for use with mechanical handling equipment suited to the weight of the unit, and positioned to evenly support it.

3.3.8.1 Pumping assembly. The pump shall have a minimum flow capacity of 50 GPM. The pump shall conform to MIL-PRF-52109, with the exception that the pump shall be driven by an electric motor. The input voltage shall be 24 volts, direct current. The motor armature brushes shall have a minimum average operational life of 1000 hours. The motor, complete with all receptacles and connectors, shall be explosion-proof in accordance with the National Electrical

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Code NFPA No. 70, Class 1, Division 1, Group D; and splash-proof in accordance with NEMA-MG-1.

3.3.8.2 Filter/Separator. The pump unit shall have an in-line fuel filter/separator conforming to MIL-PRF-52666. The filter/separator's throughput shall be rated at not less than 50 GPM.

3.3.8.3 Dispensing assembly. The pump unit shall have dispensing equipment consisting of the necessary hoses, fittings, and nozzles for two independent dispensing points. The hoses shall be reinforced, non-collapsible, lightweight, high pressure, 40-feet (long) x 1.50-inches (ID), and shall conform to MIL-PRF-370. The fittings shall be quick-disconnect couplings conforming to A-A-59326. The automatic shutoff nozzles shall conform to A-A-52030, type II, size 2.

3.3.8.4 Hose reel. Each dispensing point shall have a reel, or suitable device, with selectable automatic stop positions and a self-retracting function, to stow and deploy the hose.

3.3.8.5. Power switch. The pump unit shall have an ON/OFF power switch at the pump. In addition, the unit shall have a remote control capable of switching the pump ON/OFF from a location up to forty feet from the pump. The remote ON/OFF power switching device shall not interfere with the operation of the pump unit, or other electrical or transceiver equipment.

3.3.8.6 Power cable receptacle. The pump shall have an easily accessible receptacle interface that conforms to MIL-PRF-62122 (NATO STANAG No. 4074) for connection to a remote power supply.

3.3.8.7 Grounding device. The type I pump unit shall be equipped with a grounding rod, static ground cable with attachment devices, and a static discharge reel. The reel shall have a self-retracting function to stow and deploy the grounding cable.

3.3.8.8. Flow meter. When specified (see 6.2), the type I pump unit shall have a flow meter to measure the quantity of fuel pumped, in gallons.

3.3.9 Storage provisions. Unless otherwise specified (see 6.2), provisions shall be made for the storage of the fuel nozzles, replacement parts, additional consumables, and maintenance and operation manuals. A weather resistant compartment with 24.00-inch x 16.00-inch x 7.00-inch (high), nominal internal dimensions shall be provided at a minimum.

3.4 Performance characteristics.

3.4.1 Transportability. Under all load capacities, the complete unit shall be transportable, assembled on the host vehicle, under the following conditions without malfunction, permanent deformation, or damage to the tank, pump, or components:

a. Over improved (paved) roads at an average speed of 50 miles per hour (mph).
b. Over unimproved (unpaved) roads at an average speed of 20 mph.
c. Over cross-country terrain at an average speed of 15 mph.

3.4.2 Slope capability. Under all load conditions, the complete unit shall be transportable, assembled on the host vehicle, along slopes of up to 20 percent and up/down-grade on a ramp of a 30 percent slope, without damage to the tiedown or mounting provisions. Neither traverse shall degrade operation of the pump at rated capacity, on a level surface.

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3.4.3 Emergency stop. The complete unit shall withstand the effects of emergency stops at forward speeds up to 55 mph and at reverse speeds up to 10 mph, without damage to the tiedown or mounting provisions. Emergency stops shall not degrade operation of the pump at rated capacity.

3.4.4 Leakage. The tank and pump unit, complete with components, shall not leak or seep in operation under any conditions specified.

3.4.5 Hydrostatic pressure. The tank and pump units shall withstand a hydrostatic pressure of not less than 50 psi, without leakage through any component or connection.

3.4.6 Electromagnetic interference. The electromagnetic interference characteristics of the tank and pump unit power leads shall have a conducted susceptibility range of 30 Hz to 50 kHz when tested as specified.

3.5 Environmental.

3.5.1 Operating temperature. The complete tank and pump unit shall perform as specified in any ambient temperature from -25 °F to 140 °F.

3.5.2 Storage temperature. The complete unit shall not be damaged by storage in any ambient temperature from -50 °F to 160 °F.

3.6 Interface.

3.6.1 Mounting. The tank and pump unit shall be mounted on skids and have provisions for handling by forklift. Pockets shall be adequate for use with mechanical handling equipment suited to the weight of the unit, and positioned to evenly support it. Ancillary items required to ensure stability of the mounted tank, or alignment with other tanks and the pumping unit and their stability, shall be provided.

3.6.1.1 Government furnished property. When requested, the following property as applicable to contract or order, in the quantities indicated, will be loaned or made available to the manufacturer by the Government (see 6.5):

Unit	Vehicle	Qty
Type I	Truck, 5 ton, 6x6, cargo; M809 or M39 series.	1
Type II	Trailer, 5 ton, flatbed; XM1061 or M1061 series.	1

3.6.2 Tiedown/mounting provisions. The type I and type II units shall be provided with integral provisions and hardware to permit mounting and tiedown to the vehicle floor, deck, or rails of the transportation medium. The provisions shall be in accordance with MIL-STD-209, class 2. For the type I unit, interlocking of the major components is permissible to meet the tiedown requirements of MIL-STD-209. Plug and socket interlocks, or other method that does not require the use of hardware, or other mechanical attachments, shall be used.

3.6.3 Lifting provisions. The tank and pump unit shall have provisions for vertical lifting to facilitate the loading and unloading from the host vehicles. The provisions shall be in accordance with MIL-STD-209, class 1.

3.7 Support and ownership requirements.

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3.7.1 Maintainability. The type I unit, type II unit, and all installed attachments, shall be accessible for service, repair, and replacement without the removal of major assemblies or installed attachments.

3.7.2 Maintenance ratio. The tank and pump units shall have a maintenance ratio (see 6.5.1) not greater than 0.08.

3.7.3 Reliability. The specified mean-time-between-failure (MTBF) (see 6.5.2) for the tank and pump units, at rated capacity, shall be 114 hours.

3.7.4 Safety. All moving parts of the pump unit shall be guarded against hazard to personnel or damage to equipment. All electrical terminals shall be completely enclosed or insulated to prevent inadvertent contact by personnel and guarded against equipment that may cause arching. Fuel tanks, or shielding, shall be located in a manner that will not allow spills or overflows to run onto electrical equipment or promote combustion.

3.7.4.1 Electrical shock. Electrical equipment shall be effectively guarded and grounded to protect personnel from an electrical shock hazard.

3.7.4.2 Continuity. The pump unit shall have a continuous electrical circuit between the nozzle bonding wire, tank, pump unit, and the grounding rod, with a resistance not greater than 10 ohms.

3.7.5 Marking.

3.7.5.1 Identification marking. Identification markings shall be permanent, legible, and shall include the manufacturer’s identification code, the manufacturer’s part number the part or identification number (PIN), and the national stock number (NSN).

3.7.5.2 Instruction plates. Each tank and pump unit shall be equipped with instruction plates, including diagrams, warnings and cautions, and procedures to be followed in assembling, operating, or servicing the units. The instruction plates shall be permanently attached in conspicuous, protected locations.

3.7.6 Materials. The materials selected for construction are the responsibility of the contractor and shall be subject to all provisions of this specification. The materials shall be of sufficient durability to meet all the performance requirements as specified herein. The material shall have no adverse effect on the health of the personnel when used for its intended purpose. Whenever feasible, toxic chemicals, hazardous substances, or ozone depleting chemicals (ODC) shall not be used.

3.7.6.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.7.6.2 Deterioration prevention and control. The unit shall be fabricated from compatible materials, inherently corrosion resistant, or treated to provide protection against the various forms of corrosion and deterioration to which the material is susceptible. Dissimilar metals shall not be

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used in intimate contact unless protected against galvanic corrosion. The materials of the tanks and pump unit shall be compatible with all types of motor gas, diesel fuel, and aviation fuel.

3.7.6.3 Cleaning, treatment, and painting. All surfaces, normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard commercial practice. Special color, pattern, or requirements shall be specified in contract or order (see 6.2).

4. VERIFICATIONS

4.1 Classification of inspections. Inspections are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified (see 6.2) a first article inspection shall be performed on a complete unit(s). This inspection shall include the examination of 4.4 and the tests of 4.5.

4.3 Conformance inspection. Production unit inspection shall include examination of 4.4 and the testing of 4.5.9.1.

4.4 Examination. Each unit shall be physically examined for compliance with the requirements indicated by an "X" in Table II. This element of inspection shall encompass all examinations of performance, safety, human engineering, and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects lessening required efficiency shall constitute cause for rejection.

TABLE II. Examinations.

Item	Type I	Type II	Examination	Reference paragraph	Method of examination
101	X		Type I tanks and pump unit physical characteristics not as specified.	3.3.1	Visual/Doc.
102		X	Type II tank and pump unit physical characteristics not as specified	3.3.2	Visual/Doc.
103	X	X	Tank not as specified.	3.3.3	Visual/Doc./SIE
104	X	X	Bottom loading valve not as specified.	3.3.4	Visual/Doc./SIE
105	X		Fuel sensor cutoff not as specified.	3.3.5	Visual/Doc.
106	X	X	Tank access not as specified.	3.3.6	Visual/SIE
107	X	X	Manifold not as specified.	3.3.7	Visual/Doc.
108	X		Pump unit not as specified.	3.3.8	Visual/Doc.
109	X	X	Lifting provisions not as specified.	3.6.1	Visual/Doc.
110	X	X	Tiedown/mounting provisions not as specified.	3.6.2	Visual/Doc./SIE
111	X	X	Maintainability not as specified.	3.7.1	Visual/SIE

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TABLE II. Examinations - Continued

Item	Type I	Type II	Examination	Reference paragraph	Method of examination
112	X	X	Marking not as specified.	3.7.5	Visual
113	X	X	Materials not as specified.	3.7.6	Visual/Doc.
114	X	X	Cleaning, treatment and painting not as specified.	3.7.6.3	Visual/Doc.

Doc. = Document, SIE = Standard Inspection Equipment

4.5 Tests. First article inspection shall be performed as listed in Table III.

4.5.1 Test conditions. All performance data shall be corrected to standard sea level conditions of barometric pressure of 29.92-inches of mercury, liquid fuel at 60 °F having a specific gravity of 0.75 and vapor pressure of 4.0 psi. Water at a temperature of between 60 °F to 80 °F may be used as the test fluid. For testing, the dispensing unit shall be mounted on the vehicle and secured with tie downs.

TABLE III. Tests.

Number	Test	Requirement paragraph	Test paragraph
101	Transportability	3.4.1	4.5.2
102	Slope	3.4.2	4.5.3
103	Emergency stop	3.4.3	4.5.4
104	Leakage	3.4.4	4.5.5
105	Hydrostatic pressure	3.4.5	4.5.6.1
106	Electromagnetic Interference	3.4.6	4.5.6.2
107	Operating Temperature	3.5.1	4.5.7.1
108	Storage Temperature	3.5.2	4.5.7.2
109	Maintenance ratio	3.7.2	4.5.8.1
110	Reliability	3.7.3	4.5.8.2
111	Safety	3.7.4	4.5.9

4.5.2 Transportability. The type I unit shall be driven and the type II shall be towed over a smooth, improved (paved) road at an average speed of 50 mph for 100 miles; over unimproved (unpaved) roads at an average speed of 20 mph for a total of 100 miles; and cross country at an average speed of 15 mph for a total of 100 miles. The tanks shall be full for 50 percent, half full for 25 percent, and empty for 25 percent of the mileage on each type of surface. After each of the above segments of the test, the units shall be operated at rated capacity for 1 hour. Permanent deformation, damage to components of the unit, or failure to operate as prescribed shall constitute failure of this test.

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4.5.3 Slope. The fully loaded units shall be moved along a side slope of 20 percent. The procedure shall be repeated in the opposite direction. The truck shall be driven up-grade on a ramp of 30 percent slope, then down-grade. After negotiating the ramps, the units shall be operated at rated capacity for not less than 1 hour. Permanent deformation or damage to components of the unit or failure to operate as prescribed shall constitute failure of this test.

4.5.4 Emergency stop. The type I unit shall be subjected to a series of emergency stops while fully loaded. The truck shall be stopped while moving forward at 10 mph, then five consecutive emergency stops from each of the following speeds: 20, 30, and 45 mph forward, and finally 10 mph in reverse. The type I unit shall be operated at rated capacity for not less than 1 hour on completing the emergency stops. The fully loaded type II unit shall be subjected to a series of emergency stops following the same procedure as for the type I unit, except that the emergency stops shall be from 10, 20, and 35 mph in the forward direction and at 5 mph in reverse; five consecutive times from each speed. The mounting and tiedown systems shall be inspected throughout both tests. Permanent deformation or damage to components of the unit or failure to operate as prescribed herein shall constitute failure of this test.

4.5.5 Tank leakage. The interior and exterior of the tank shall be completely cleaned of foreign matter and dried. Once the tank is completely clean and dry, close all openings such as valves and fill cap. Plug the fill cap vent, and subject the tank to a constant internal air pressure of 3.0 to 5.0 psig. Spray liquid detergent over the entire exterior surface of the tank at a pressure of not less than 70.0 psig. Applying the liquid detergent by brush or other means shall not be acceptable. Observe the exterior of the tank. Leaks shall be shown by fizzing or bubbling of the liquid detergent. Surface rupture, leakage, deformation, or malfunctioning hardware shall constitute failure of this test.

4.5.6 Pump unit. To include hoses, couplings, nozzles, valves, and filter separator.

4.5.6.1 Hydrostatic pressure. The pump unit shall be subjected to a hydrostatic pressure of not less than 50 psi for 20 minutes. Any leakage, rupture, loss of pressure, permanent deformation or damage to components shall constitute failure of this test.

4.5.6.2 Electromagnetic interference. The pump unit power leads shall be tested using the CS101 method in MIL-STD-462. Test results not within the range of 30 Hz to 50 kHz shall constitute failure of this test.

4.5.7 Environmental.

4.5.7.1 Operating temperature. A tank and pump unit shall be placed in a temperature-controlled chamber. The chamber temperature shall be varied by hour to cycle through the temperatures from -25 °F to 140 °F in a 24-hour period. Seven 24-hour cycles shall be run to complete the test. The unit shall be examined before the test start and the condition of all items noted. During the final cycle at -25 °F and 140 °F the pump unit shall be examined and placed in operation for a 30-minute period. Permanent deformation or damage to components of the unit or failure to operate as prescribed herein shall constitute failure of this test.

4.5.7.2 Storage temperature. A tank and pump unit shall be placed in a temperature-controlled chamber. At a rate not to exceed 6 °F per minute, the chamber temperature shall be

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lowered to -50 °F. When this temperature is reached and the unit temperature has stabilized, maintain the temperature for four hours. At the end of the four hours, the chamber temperature shall be raised, at a rate not to exceed 6 °F per minute, to -25 °F. At this temperature, the pump unit shall be placed in operation for 30 minutes. The test shall continue by raising the chamber temperature at a rate not to exceed 6 °F per minute to 160 °F, stabilize the temperature for four hours, at a rate of 6 °F per minute lower the temperature to 140 °F. At this temperature, place the unit in operation for 30 minutes. Permanent deformation or damage to components of the unit or failure to operate as prescribed herein shall constitute failure of this test.

4.5.8 Support and ownership requirements.

4.5.8.1 Maintenance ratio. The unit shall be operated as specified herein for a minimum of 100 hours. Operating data accumulated during previous testing shall be included in data for reliability calculations. A calculated maintenance ratio (see 6.6.1) greater than 0.08 shall constitute failure of this test.

4.5.8.2 Reliability. The unit shall be operated as specified herein for a minimum of 100 hours. Operating time accumulated during previous testing shall be included in data for reliability calculations. A calculated MTBF (see 6.6.2) less than 114 hours shall constitute failure of this test.

4.5.9 Safety. As part of the functional test all aspects of operation shall be evaluated for safety of personnel and protection of equipment. Evaluation shall include, but is not limited to, the requirements specified in 3.7.4. All observed unsafe conditions that result from design, construction, or operation shall be noted. Any observed infringement shall constitute failure of this test.

4.5.9.1 Continuity. The ground wire on the pump unit and each nozzle bonding wire shall be connected to an ohmmeter to determine continuity. An ohmmeter reading greater than 10 ohms shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The type I dual tank and pump unit is intended to be mounted onto the flatbed of a truck, and the type II single tank and manifold are intended to be mounted on a trailer.

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Both types dispense liquid petroleum fuels. The units are military unique in that no commercial equivalent exists in this capacity for operation in an extreme environment, under conditions typically found on the battlefield.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of fuel dispensing unit required (see 1.2).
c. Issue of DoDISS to be cited in solicitation, and, if required, the specific issue of individual documents referenced (see 2.2).
d. When a first article inspection is required (see 3.2 and 4.2).
e. When a secondary pilot valve and second level sensor are required (see 3.3.5.3).
f. When a flow meter is required (see 3.3.8.8).
g. When storage provisions are not required (see 3.3.9).
h. When special cleaning, treatment, or painting is required (see 3.7.6.3).
i. Packaging requirements (see 5.1).

6.3 First Article. When a first article inspection is required, the item(s) should be a first article production unit. The contracting officer should include specific instructions in acquisitions' documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first article(s).

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manuals to be furnished with each pump and tank unit. If manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12 L, Acquisition Management Systems and Data Requirements Control List (AMSDL) should be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The manuals should be acquired under a separate contract line item in the contract.

6.5 Government furnished property. The contracting officer should arrange to loan, or provide access to, the property specified in 3.6.1.1, as applicable, when required.

6.6 Definitions.

6.6.1 Maintenance ratio. The maintenance ratio is defined as the ratio of the total active maintenance in man-hours required (scheduled and unscheduled) to the total operating time.

6.6.2 Mean-time-between-failure. The mean-time between failure (MTBF) is the ratio of the total operating time to the total number of failures.

6.7 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

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This example describes a part numbering system for specification MIL-PRF-40136K:

M40136-2
- Type I
- Type II

6.8 Subject term (key word) listing.

Bottom loading
Fuel service
Fuel storage
Fuel transport

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – AT
Air Force – 99

Preparing activity:
Army – AT

Review activities:
DLA – CC

Project (4930-0044)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-40136K	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
Tank and Pump Unit, Fuel Dispensing, for Truck Mounting, and Single Tank, Fuel Dispensing, 525 Gallon, Trailer Mounted

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USA TACOM, AMSTA-TR-D/210 (FWH)	b. TELEPHONE (Including Area Code) (1) Commercial (810)574-7744 (2) DSN 786-7744
c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command Attn: AMSTA-TR-D/210 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 681-9340 DSN 761-9340

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